PART-1 : PHYSICS

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

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1. A potentiometer wire has length L and the emf of the driver cell is E_0 with negligible internal resistance. With a cell of emf E, the balance length is $\frac{L}{4}$. If the length of the potentiometer wire is doubled (without changing its diameter), the new balance length with the same cell will be

(A) $\frac{L}{4}$ (B) $\frac{L}{3}$

(C) $\frac{L}{2}$ (D) $\frac{2L}{3}$

Ans. C

Sol. In the first case, the drop of potential per unit length of the potential wire is $\frac{E_0}{L}$. Hence

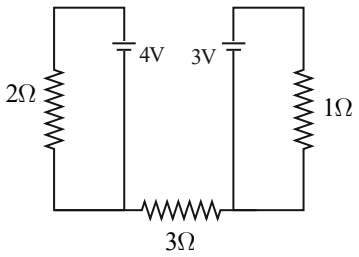
$$E=\left(\frac{L}{4}\right)\times\frac{E_0}{L}=\frac{E_0}{4} \qquad ..(i)$$

In the second case, the potential drop per unit length = $E_0/2L$. If x is the balance length now, then

$$E=\frac{xE_0}{2L} \qquad ..(ii)$$

Equ. (i) & (ii) we get $x=\frac{L}{2}$.

2. The potential difference through the 3Ω resistor shown in figure :



(A) Zero (B) 1 V

(C) 3.5 V (D) 7 V

Ans. A

Sol. The two sub circuits are closed loops. They cannot send any current through the 3Ω resistor. Hence the potential difference across the 3Ω resistor is zero.

3. A cylindrical block of wood floats vertically with 80% of its volume immersed in a liquid at 0°C. When the temperature of the liquid is raised to 62.5°C, the block just sinks in the liquid. The coefficient of cubical expansion of liquid is

(A) $1\times10^{-3}\,K^{-1}$ (B) $2\times10^{-3}\,K^{-1}$

(C) $3\times10^{-3}\,K^{-1}$ (D) $4\times10^{-3}\,K^{-1}$

Ans. D

Sol. Weight of cylinder = weight of the liquid displaced,

i.e. $Ah\rho_b = A\times(0.8h)\times\rho_0$ or $\rho_0 = \frac{\rho_b}{0.8} = \frac{5\rho_b}{4}$ .....(i)

Here ρ_0 is the density of the liquid at 0°C. For the block to sink in the liquid at t°C, the density of the liquid must change from ρ_0 to ρ at t°C. Now

$$\rho = \frac{\rho_0}{1+\gamma t} \qquad .....(ii)$$

For the block to sink $\rho = \rho_b$. Using (i) and (ii), we have

$$\rho_b = \frac{5\rho_b/4}{1+\gamma t} \quad \text{or } 1+\gamma t = \frac{5}{4}$$

Give $t = 62.5°C$. Using this value, we get

$\gamma = 4\times10^{-3}\,(°C)^{-1}$ or K^{-1}

4. A metal ball immersed in alcohol weighs W_1 at 0°C and W_2 at 59°C. The coefficient of cubical expansion of metal is equal than that of alcohol. If the density of the metal is large compared to that of alcohol, then

(A) $W_1 > W_2$ (B) $W_1 = W_2$

(C) $W_1 < W_2$ (D) $W_2 = \frac{W_1}{2}$

Ans. B

Sol. Let V_0 be the volume of the metal at 0°C and V_t its volume at t°C. At temperature t the upthrust is

$U_t = V_t\rho_t g$

where ρ_t is the density of alcohol at temperature t.

Now $V_t = V_0(1+\gamma t)$

where γ is the coefficeint of cubical expansion of alcohol and V_0 is the volume of alcohol displaced at temperature t = 0°C. Now the density of alcohol at temperature t is

$$\rho_t = \frac{\rho_0}{1+\gamma t}$$

Where ρ_0 is the density of alcohol at t = 0°C.

Hence $U_t = V_0(1+\gamma t)\times\frac{\rho_0 g}{(1+\gamma t)} = V_0\rho_0 g = U_0$

where U_0 is the upthrust at 0°C. Since the upthrust is independent of temperature, $W_1 = W_2$.

5. Two objects of masses m and 4m are at rest at infinite separation. They move towards each other under mutual gravitational attraction. Then, at a separation r, which of the following is true?

(A) The total kinetic energy of the system is not zero.

(B) The force between them is not zero.

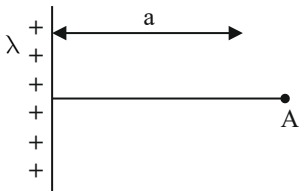
(C) The centre of mass of the system is at rest.

(D) All the above are true.

Ans. D

Sol. At a finite separation, the total kinetic energy of the system of two masses and the force between them are both finite. Since the two masses are at rest initially and there is no external force, the centre of mass cannot move.

6. A point charge +q is projected from point A towards an infinitely long line charge having linear charge density λ with kinetic energy K_0. The distance of the closest approach will be :



(A) $ae^{-\left(\frac{2\pi\varepsilon_0 K_0}{q\lambda}\right)}$

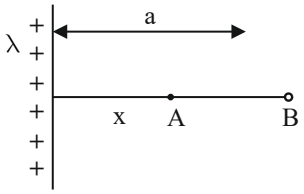
(B) $ae^{-\left(\frac{\pi\varepsilon_0 K_0}{q\lambda}\right)}$

(C) $a\left(1-\frac{2\pi\varepsilon_0 K_0}{q\lambda}\right)$

(D) $\left(\frac{2\pi\varepsilon_0 K_0}{q\lambda}\right)$

Ans. A

Sol. Work done by electric field during motion of charge +q from B to A ('x' is distance of closest approach)



$$q(V_A - V_B) = -q.\int_a^x \frac{\lambda}{2\pi\varepsilon_0 x}dx \qquad \Rightarrow W = q\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{a}{x}$$

By work energy theorem

$$\Rightarrow \qquad K_0 = \frac{q\lambda}{2\pi\varepsilon_0 x}\ln\frac{a}{x} \qquad \Rightarrow x = ae^{-2\pi\varepsilon_0 K_0/\lambda q}$$

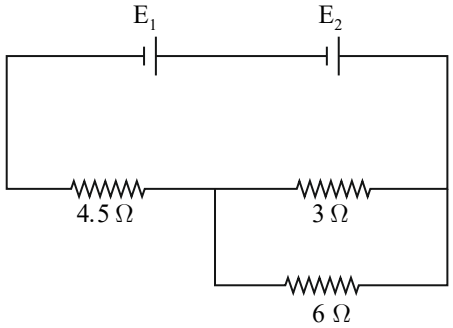
SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

7. In the circuit shown in figure, cells E_1 and E_2 have emfs 4 V and 8 V and internal resistances $0.5\,\Omega$ and 1Ω respectively.



(A) The potential difference across E_1 is 4.25 V (B) The potential difference across E_1 is 3.75 V

(C) The potential difference across E_2 is 8.55 V (D) The potential difference across E_2 is 7.5 V

Ans. AD

Sol. Equivalent resistance of the parallel combination of $3\,\Omega$ and 6Ω is $R=\frac{3\times6}{(3+6)}=2\Omega$. As the cells are in opposition, net emf $E=E_2-E_1=8-4=4V$.

Therefore, current is

$$I=\frac{E}{R+4.5+r_1+r_2}=\frac{4}{2+4.5+0.5+1}=0.5A$$

Potential difference across E_1 is $V_1=E_1+Ir_1=4+0.5\times0.5=4.25V$

Potential difference across E_2 is $V_2=E_2-Ir_2=8-0.5\times1=7.5V$

Hence the correct choices are (A) and (D).

8. If three bulbs of 40 W, 60 W and 100 W are connected in series with a 200 V power supply, then

(A) The potential difference will be maximum across the 40 W bulb

(B) 100 W bulb will glow brightest

(C) The 40 W bulb has the maximum resistance

(D) The current in the circuit is 1.2 A

Ans. AC

Sol. Resistance of 40 W bulb $= \frac{V^2}{W} = \frac{200 \times 200}{40} = 1000\Omega$

Resistance of 60 W bulb $= \frac{200 \times 200}{60} = 666.7\ \Omega$

Resistance of 100 W bulb $= \frac{200 \times 200}{100} = 400\ \Omega$

Hence choice (C) is correct. Since the bulbs are connected in series, the current in each is the same. Hence choice (B) is incorrect. The total resistance = 1000 + 666.7 + 400 = 2066.7Ω. Therefore, the current in the circuit is

$$I = \frac{200}{2066.7} = 0.097A$$

Hence the choice (D) is also incorrect. Since the 40 W bulb has the highest resistance and the current in each bulb is the same, the potential difference across the 40 W bulb is the highest. Hence the correct choices are (A) and (C).

9. A satellite of mass m is moving in a circular orbit of radius r around a planet of mass M

(A) The magnitude of angular momentum with respect to the centre of the orbit is $m\sqrt{GMr}$, where G is the gravitation constant.

(B) The magnitude of the angular momentum is $mR\sqrt{2gr}$ where g is the acceleration due to gravity on the surface of the planet.

(C) The direction of angular momentum is parallel to the plane of the orbit.

(D) The direction of angular momentum is perpendicular to the plane of the orbit.

Ans. AD

Sol. At a certain instant of time let r be the radius vector of the satellite from the centre of its circular orbit. If the velocity of the satellite is υ its angular momentum is given by

$$L = mr\upsilon \qquad .....(i)$$

The gravitational force of attraction on the satellite is GMm/r^2 which provides the necessary centripetal force $\frac{GMm}{r^2} = \frac{m\upsilon^2}{r}$ or $(mr\upsilon) = m\sqrt{GMr}$ .....(ii)

From Eqs. (i) and (ii), we have $L = m\sqrt{GMr}$

This gives the magnitude of angular momentum. The direction of angular momentum is perpendicular to the plane of the orbit. The correct choices are (A) and (D).

10. A positive charge q is fixed at the origin. An electric dipole with dipole moment $\vec{p}$ is placed along the x-axis far away from the origin with $\vec{p}$ pointing along the positive x-axis. The kinetic energy when it reaches a distance x from the origin is K and the magnitude of the force experienced by charge q at this moment is F. Then

(A) K veries as $\frac{1}{x}$ (B) K veries as $\frac{1}{x^2}$

(C) F veries as $\frac{1}{x^2}$ (D) F veries as $\frac{1}{x^3}$

Ans. BD

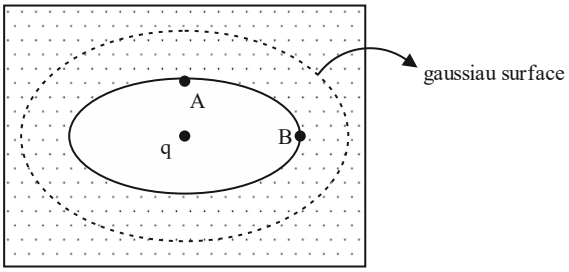
Sol. Potential energy at position x is $U = -\frac{1}{4\pi\varepsilon_0}.\frac{qp}{x^2}$

Potential energy at position x is $(x \to \infty), U = 0.$

Hence K = change in P.E. $= 0 - \left[-\frac{1}{4\pi\varepsilon_0}\frac{q\,p}{x^2}\right] = \frac{1}{4\pi\varepsilon_0}.\frac{q\,p}{x^2}$ $\Rightarrow F = \frac{dU}{dx} = -\frac{q\,p}{4\pi\varepsilon_0 x^3}$

Hence the correct choices are (B) and (D).

11. An ellipsoidal cavity is made within a perfect conductor. A positive charge q is placed in the cavity and A and B are two points on the surface of the cavity as shown in figure. Then



(A) Electric field near A in the cavity = electric field near B in the cavity.

(B) Charge density at A < charge density at B

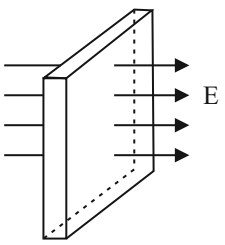
(C) Potential at A = potential at B

(D) Total electric flux through the gaussian surface shown is $\frac{q}{\epsilon_0}$

Ans. BC

Sol. The surface of a perfect conductor is an equipo-tential surface. Charge density at B is greater than at A because curvature at B is greater than that at A. From Gauss's theorem, flux through gaussian surface will be zero as there is zero electric field at gaussian surface

12. An uncharged conducting thin square plate of side 1 m is placed in an uniform electric field of magnitude 200N/C as shown in the figure. The field direction is perpendicular to the plane of plate. For this situation, mark out the correct statement (s). [In the options only big faces have been considered]:



(A) The net charge on each face of the plate is zero.

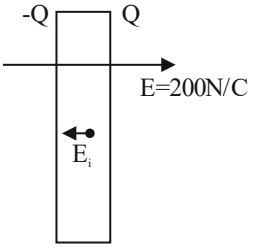
(B) The net charge on each face of the plate is non-zero and having the magnitude equal to 1.77×10^{-9} C.

(C) The net electric field intensity inside the plate is zero.

(D) The electric field intensity at the surface of plate is discontinuous while potential is continuous.

Ans. BCD

Sol. As in electrostatics, the electric field inside the bulk of the material of the conductor is zero. So some charge gets induced on two faces of the plate to make electric field zero inside the conductor.



Let Q and -Q appears on right and left face as shown, then induced electric field,

$$E_i = \frac{Q}{2\varepsilon_0 A} + \frac{Q}{2\varepsilon_0 A} = \frac{Q}{\varepsilon_0 A} \text{ (towards left)}$$

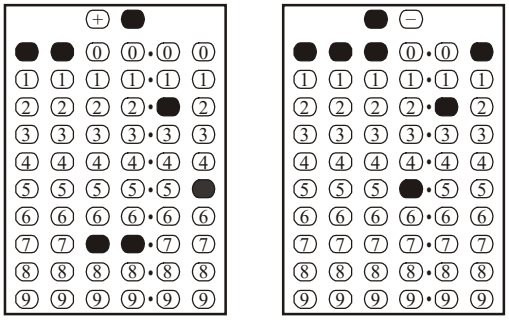
For $\vec{E}$ inside plate to be zero

$$E_i = E \qquad \Rightarrow Q = \varepsilon_0 A \times 200 = 1.77\times10^{-9}\,C$$

The electric field is discontinuous at surface but potential is continuous.

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

13. A body is dropped from a height h equal to $\frac{R}{2}$, where R is the radius of the earth. If it will hit the surface of the earth with a speed $\upsilon = a\upsilon_e$, where υ_e is the escape velocity from the surface of the earth. Find a

Ans. 0.57 or 0.58

Sol. Total energy of the body at height h is

$$E_i = -\frac{GmM}{(R+h)} \qquad .....(i)$$

Total energy when it hits the surface of the earth is

$$E_f = \frac{1}{2}m\upsilon^2 - \frac{GmM}{R} \qquad .....(ii)$$

From conservation of energy, $E_i = E_f$, i.e.

$$-\frac{GmM}{(R+h)} = \frac{1}{2}m\upsilon^2 - \frac{GmM}{R} \qquad \Rightarrow \upsilon = \sqrt{\frac{2GM}{R}} \times \sqrt{\frac{h}{(R+h)}}$$

$$= \upsilon_e\sqrt{\frac{R}{(R+h)}} \Rightarrow \text{For } h = \frac{R}{2}, \upsilon = \frac{\upsilon_e}{\sqrt{3}} = 0.577V_e$$

14. Assuming the earth to be a sphere of uniform mass density, find the percentage decrease in the weight of a body when it is taken to the end of a tunnel 32 km below the surface of the earth. Radius of earth = 6400 km

Ans. 0.5

Sol. $g' = g\left(1-\frac{d}{R}\right) = g\left(1-\frac{32}{6400}\right) = \frac{199g}{200}$

$$\text{Decrease in weight} = mg - mg' \qquad = mg\left(1-\frac{199}{200}\right) = \frac{mg}{200}$$

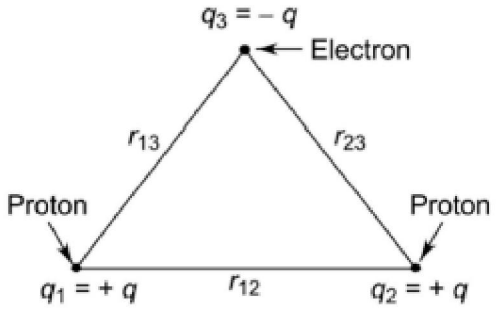
$$\text{Percentage decrease} = \frac{mg/200}{mg} \times 100 = 0.5\%$$

15. A neutral hydrogen molecule has two protons and two electrons. If one of the electrons is removed we get a hydrogen molecular ion $\left(H_2^+\right)$. In the ground state of H_2^+ the two protons are separated by roughly $1.5\ \mathring{A}$ and the electron is roughly $1\ \mathring{A}$ from each proton. What is the potential energy of the system? (in eV)

Ans. -19.2

Sol. Refer to figure. The total potential energy of the arrangement of charges is the sum of the energies of each pair of charges. The potential energy of the system comprising the three charges q_1, q_2 and q_3 is

$$U = W_1 + W_2 + W_3 \qquad = \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1q_2}{r_{12}} + \frac{q_1q_3}{r_{13}} + \frac{q_2q_3}{r_{23}}\right)$$



Here $q_1 = q_2 = q = +1.6\times10^{-19}\,C(\text{proton}),\ q_3 = -q = -1.6\times10^{-19}\,C(\text{electron}),$

$r_{12} = 1.5\,\mathring{A} = 1.5\times10^{-10}\,m, r_{23} = 1\,\mathring{A} = 1\times10^{-10}\,m$ and $\dfrac{1}{4\pi\varepsilon_0} = 9\times10^9\,Nm^2C^{-2}.$

Thus $U = -\dfrac{4}{3}.\dfrac{q^2\times10^{10}}{4\pi\varepsilon_0}\,\text{joule} = -\dfrac{4}{3}.\dfrac{q\times10^{10}}{4\pi\varepsilon_0}\,eV$

$= -\dfrac{4\times1.6\times10^{-19}\times10^{10}\times9\times10^9}{3} = -19.2eV$

16. Two isolated metal spheres of radii R and 2R are charged such that both have the same surface charge density σ. The spheres are located far away from each other. When they are connected by a thin conducting wire, the new surface charge density on the bigger sphere will be $a\sigma$ then find a.

Ans. 0.83

Sol. Initial charge on sphere of radius R is $Q_1 = 4\pi R^2\sigma$ and on sphere of radius 2R is $Q_2 = 4\pi(2R)^2\,\sigma = 16\pi R^2\sigma$

Total initial charge is $Q = Q_1 + Q_2 = 20\pi R^2\sigma$

Initial potential of the sphere of radius R is $V_1 = \dfrac{Q_1}{4\pi\varepsilon_0 R}$

and of the sphere of radius 2R is $V_2 = \dfrac{Q_2}{4\pi\varepsilon_0(2R)}$

When the spheres are connected by a thin wire, charge will flow from one sphere to the other until their potentials become equal. Let Q_1' and Q_2' be the new charges, then the potential of each sphere will be

$V = \dfrac{Q_1'}{4\pi\varepsilon_0 R} = \dfrac{Q_2'}{4\pi\varepsilon_0(2R)}$ Which gives $Q_1' = \dfrac{Q_2'}{2}$

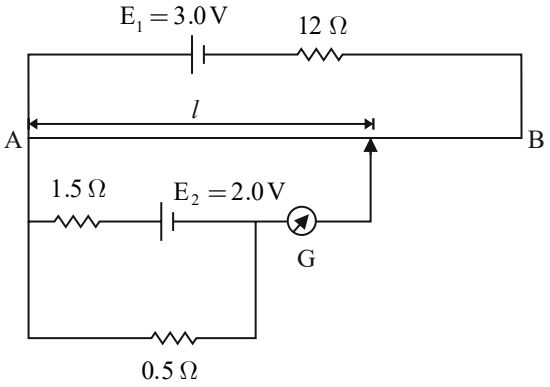
From conservation of charge, we have

$Q_1 + Q_2 = Q_1' + Q_2' \Rightarrow 20\pi R^2\sigma = Q' + Q_2' = \dfrac{3Q_2'}{2} \Rightarrow Q_2' = \dfrac{40}{3}\pi R^2\sigma$

$\therefore$ New surface charge density on the sphere of radius 2R is

$\sigma' = \dfrac{Q_2'}{4\pi(2R)^2} = \dfrac{\frac{40}{3}\pi R^2\sigma}{16\pi R^2} = \dfrac{5\sigma}{6}$

17. In the potentiometer circuit shows in figure find the value of ℓ when the galvanometer shows no deflection. The length of wire AB is 100 cm and its resistance is 3Ω (in cm)



Ans. 83.3

Sol. Current flowing in AB due to driver cell of cm 3.0 V is $I_1 = \frac{3.0}{(12+3)} = 0.2A$

$\therefore$ Potential difference across AB due to E_1, is $= 0.2 \times 3 = 0.6V$

Potential difference across I due to E_1 is (l is in cm) $V_1 = \frac{0.6}{100} \times l$ ...(i)

Current through 0.5Ω resistance due to E_2 is $I_2 = \frac{2.0}{(1.5+0.5)} = 1.0A$

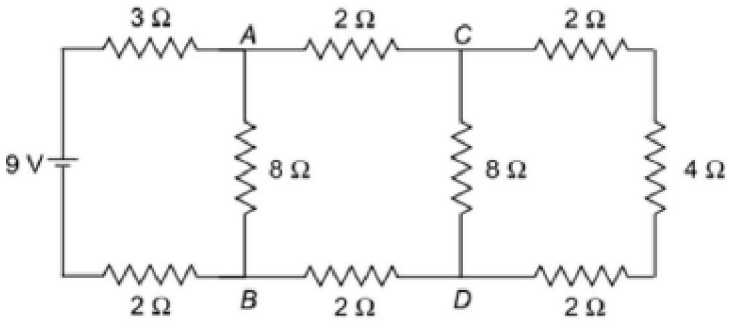
$\therefore$ Potential difference across 0.5Ω resistance $= 0.5 \times 1.0 = 0.5V.$

At balance length l, the potential difference across l due to E_2 is $V_l' = 0.5V$ .....(ii)

At balance point $V_l = V_l'$. Equation (i) and (ii), we have

$\frac{0.6}{100} \times l = 0.5 \quad \Rightarrow l = 83.3$ cm

18. In the cirucit shown figure, the current through 4Ω resistor is (in amp)



Ans. 0.25

Sol. The equivalent resistance between A and B of the circuit to the right of AB is 4Ω. Therefore, total resistance $= 3+4+2 = 9\Omega$ Current $I = \frac{9V}{9\Omega} = 1A$. The current is equally divided in the 8Ω resistor between A and B and the remainder 8Ω. Hence current in AC = 0.5 A. This current is equally divided between CD and the circuit to the right of CD. Therefore, current in the 4Ω resistor = 0.25 A.

PART-2 : CHEMISTRY

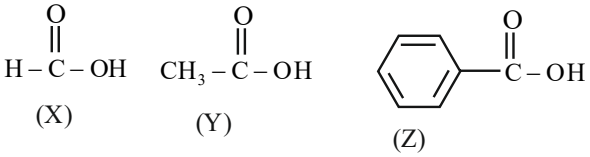
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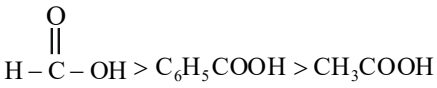
1. Arrange the following acids in decreasing order of acidity :



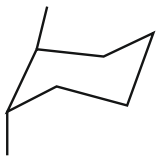
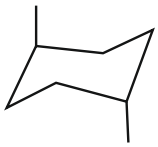
(X) (Y) (Z)

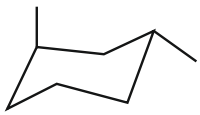
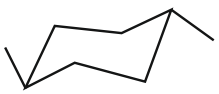
(A) X > Z > Y (B) X > Y > Z
(C) Z > X > Y (D) Z > Y > X

Ans. A

Sol. 

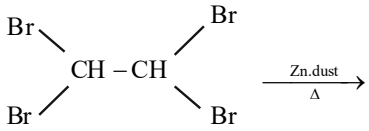
2. Which of the following compounds is most stable ?

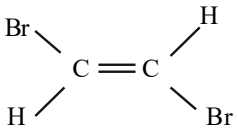
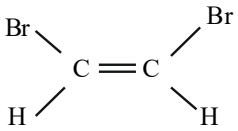
(A)  (B) 

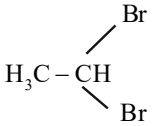
(C)  (D) 

Ans. D

Sol. Diequatorial is more stable

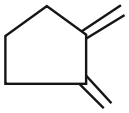
3. 

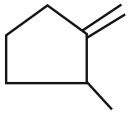
(A)  (B) 

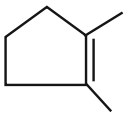
(C) $H-C\equiv C-H$ (D) 

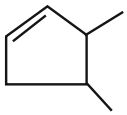
Ans. C

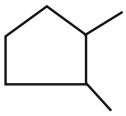
Sol. Gives elimination ; $ZnBr_2$ goes out

4.  $\xrightarrow[\Delta]{H_2(1\text{mole})}$ Product

(A) 

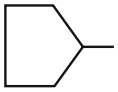
(B) 

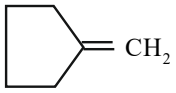
(C) 

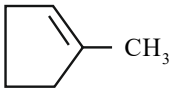
(D) 

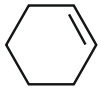
Ans. B

5. For the reaction

 $CH_2OH \xrightarrow[\Delta]{H_2SO_4}$ the major product is :

(A) 

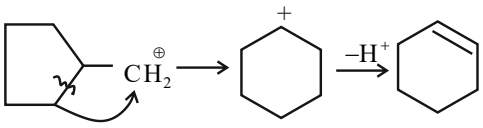
(B) 

(C) 

(D) 

Ans. C

Sol. Ring expansion :



6. The IUPAC name of $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2NH_2$ is :

(A) 2, 3 - Dimethylbutan-4-amine

(B) 2, 4--Dimethylbutan-1-amine

(C) 2,4-Dimethylbutan-4-amine

(D) 2, 3-Dimethylbutan-1-amine

Ans. D

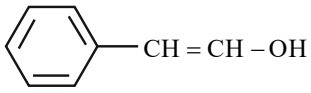
SECTION–I(ii) : (Maximum Marks : 24)

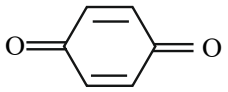
- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

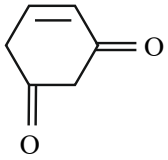
Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

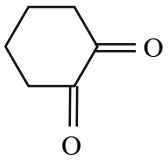
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

7. Tautomerism is exhibited by :

(A) 

(B) 

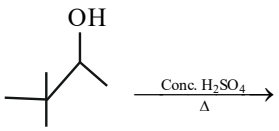
(C) 

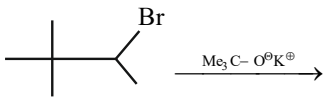
(D) 

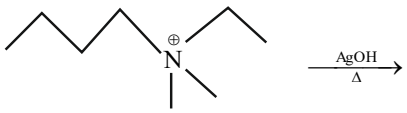
Ans. ACD

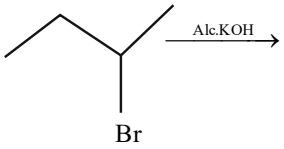
Sol. D $\Rightarrow$ Quinoid compound do not exhibit

8. Which of the following reactions will give least substituted alkene ?

(A) 

(B) 

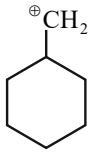
(C) 

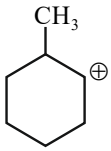
(D) 

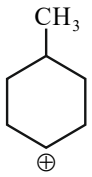
Ans. BC

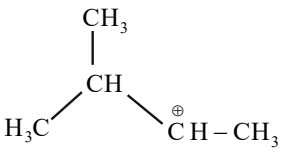
Sol. Bulky base and Bulky leaving group gives Hoffmann product

9. Which of the following carbocations would rearrange ?

(A) 

(B) 

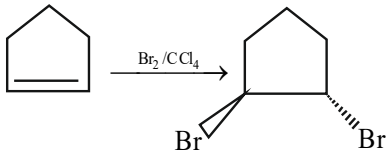
(C) 

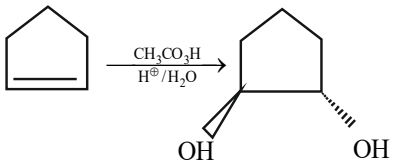
(D) 

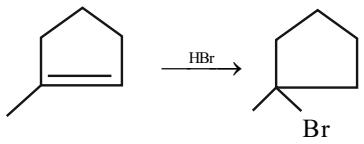
Ans. ABD

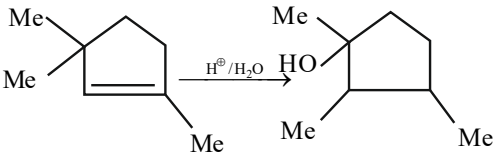
Sol. Carboncation migrates to more stable portion

10. Which of the following reactions are correct ?

(A) 

(B) 

(C) 

(D) 

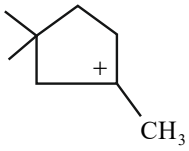
Ans. ABC

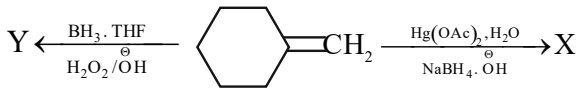
Sol. A. Anti addition

B. Anti-hydroxylation

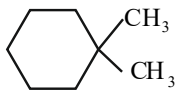
C. Markornikoff's addition

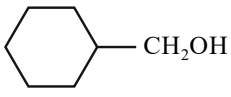
D. First form carbocation at 3°

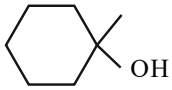


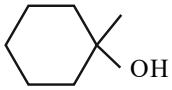
11. 

Select correct options ?

(A)  both X and Y

(B) Y is 

(C) X is 

(D)  Both X and Y

Ans. BC

Sol. A. OMDM Markovnikolf addition

B. HBO Anti-markovnikolf addition

12. $y \xleftarrow[\text{Catalyst}]{\text{Lindlar's}} R-C\equiv C-R \xrightarrow{Na/NH_3} x$

x and y are geometrical isomers $(R-CH=CH-R)$ of which type ?

(A) x is trans (B) y is cis

(C) x and y both are trans (D) x is cis, y is trans

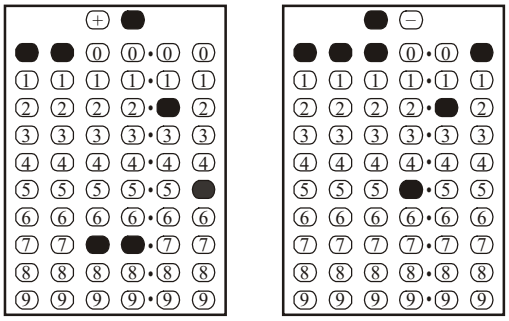
Ans. AB

Sol. x. is Anti addition

y. is syn addition

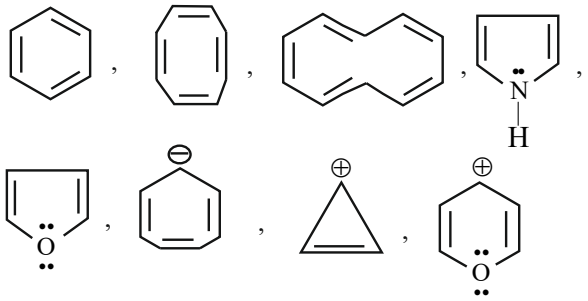
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



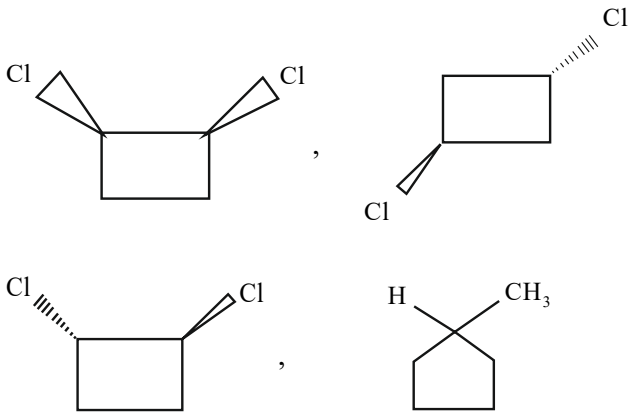
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

13. Find out number of aromatic compounds or ions from the following



Ans. 5

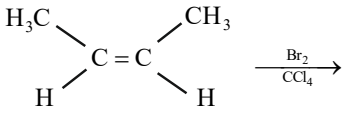
14. How many of the following do not have plane of symmetry ?

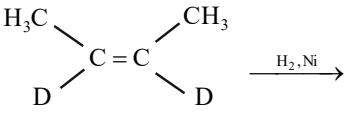


Ans. 1

Sol. C option has no plane of symmetry

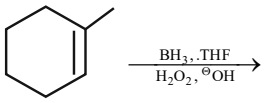
15. How many of the following addition reactions are syn addition

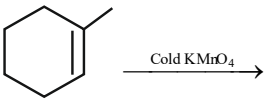
(a) 

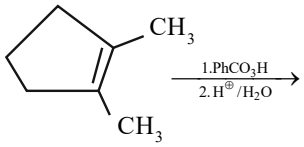
(b) 

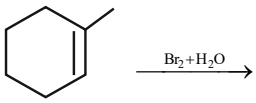
(c) $H_3C-C\equiv C-CH_3 \xrightarrow{H_2,Pd-BaSO_4}$

(d) $H_3C-C\equiv C-CH_3 \xrightarrow{Na,Liq.NH_3}$

(e) 

(f) 

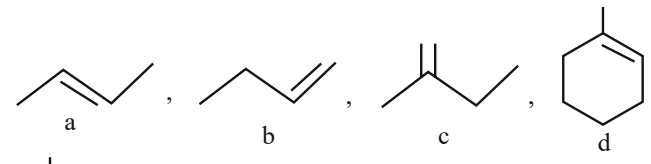
(g) 

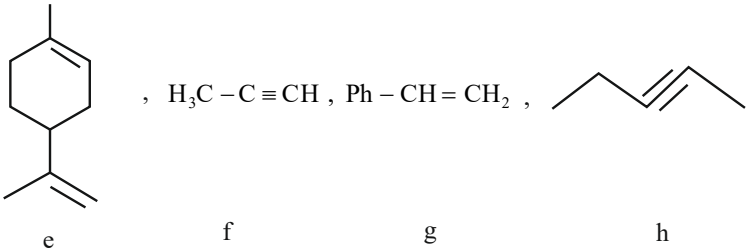
(h) 

Ans. 4

Sol. b, c, e, f $\Rightarrow$ syn addition

16. Examine the structural formulas of following compounds and find out how many compounds will produce CO_2 on oxidative ozonolysis.

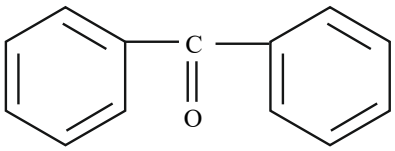




Ans. 5

Sol. b, c, e, f, g gives CO_2

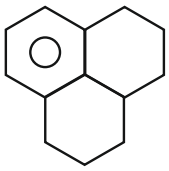
17. What is the degree of unsaturation in



Ans. 9

Sol. D.U = 9

18. Find out number of benzylic hydrogen in



Ans. 5

PART-3 : MATHEMATICS

SECTION–I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

---

1. Let $f:(1,\infty)\to(0,\infty)$ be continuous strictly decreasing function with $\lim\limits_{x\to\infty}\dfrac{f(4x)}{f(8x)}=1$. Then

$\lim\limits_{x\to\infty}\dfrac{f(6x)}{f(8x)}=$

(A) $\dfrac{1}{2}$ (B) $\dfrac{2}{3}$

(C) $\dfrac{3}{4}$ (D) 1

Ans. D

Sol. $0<f(8x)<f(6x)<f(4x)\Rightarrow 0<1<\dfrac{f(6x)}{f(8x)}<\dfrac{f(4x)}{f(8x)}$

$\Rightarrow\lim\limits_{x\to\infty}\dfrac{f(6x)}{8x}=1$

2. The value of a for which $ax^2+\sin^{-1}\left(x^2-2x+2\right)+\cos^{-1}\left(x^2-2x+2\right)=0$ has a real solution,

(A) $\frac{\pi}{2}$ (B) $\frac{-\pi}{2}$

(C) $\frac{2}{\pi}$ (D) $-\frac{2}{\pi}$

Ans. B

Sol. x = 1 (Only point in domain)

3. The equation $2\cos^{-1}x+\sin^{-1}x=\frac{11\pi}{6}$ has

(A) No solution (B) Only one solution

(C) Two solution (D) Three solution

Ans. A

Sol. $\cos^{-1}x+\left(\cos^{-1}x+\sin^{-1}x\right)=\frac{11\pi}{6}$

$\cos^{-1}x+\frac{\pi}{2}=\frac{11\pi}{6}\Rightarrow\cos^{-1}x=\frac{11\pi-3\pi}{6}$

$\cos^{-1}x=\frac{8\pi}{6}=\frac{4\pi}{3}$ which is not possible as $\cos^{-1}x\in[0,\pi]$

4. A continuous function $f(x)$ on $R\to R$ satisfies the relation $f(x)+f(2x+y)+5xy=f(3x-y)+2x^2+1\ \ \forall x,y\in R$, then f (x) is

(A) $1-x^2$ (B) $1-\frac{x^2}{2}$

(C) $1-2x^2$ (D) $2-x^2$

Ans. B

Sol. Replace y with $\frac{x}{2}$

$f(x)+f\left(\frac{5x}{2}\right)+\frac{5x^2}{2}=f\left(\frac{5x}{2}\right)+2x^2+1$

$\to f(x)=1-\frac{x^2}{2}$

5. $f(x)=-1+|x-1|,-1\le x\le 3$

$g(x)=2-|x+1|,-2\le x\le 2$

then for $x\in(0,1)$, fog (x) equals

(A) x – 1 (B) 1 – x

(C) x + 1 (D) $-(x+1)$

Ans. A

Sol. $0 < x < 1 \Rightarrow f(x) = -1-(x-1) = -x$ and $g(x) = 2-(x+1) = 1-x$

$\therefore x \in (0,1) \Rightarrow fog(x) = f(g(x))$

$= f(1-x) = -1(1-x) = x-1$

6. If $\sqrt{x+y}+\sqrt{y-x} = 5$, then $\frac{d^2y}{dx^2} =$

(A) $\frac{2}{5}$ (B) $\frac{4}{25}$

(C) $\frac{2}{25}$ (D) $\frac{1}{25}$

Ans. C

Sol. $\sqrt{x+y}+\sqrt{y-x} = 5$, $\sqrt{x+y} = 5-\sqrt{y-x}$

Sq. both sides,

$\Rightarrow x+y = 25+y-x-10\sqrt{y-x}$

$\Rightarrow 25-2x = 10\sqrt{y-x} \quad \Rightarrow -2 = \frac{10(y'-1)}{2\sqrt{y-x}}$

$\Rightarrow -2\sqrt{y-x} = 5(y'-1) \Rightarrow -\left(5-\frac{2x}{5}\right) = 5(y'-1)$

$-5+\frac{2x}{5} = 5(y'-1) \Rightarrow y'' = \frac{2}{25}$

SECTION–I(ii) : (Maximum Marks : 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

7. $\lim\limits_{n\to\infty}\left\{\left(\sqrt{5}+2\right)^n\right\}.\cos\left(\pi\left[\left(\sqrt{5}+2\right)^n\right]\right)$ is equal to

(where $n\in I^+$ and $[\bullet],\{\bullet\}$ denotes greatest integer function and fractional part function respectively)

(A) 0 when n is even
(B) –1 when n is even
(C) –1 when n is odd
(D) 0 when n is odd

Ans. BD

Sol. If n is even, then $\left[\left(\sqrt{5}+2\right)^n\right]$ is odd and $f+f'=1$

$\Rightarrow f=1-f'$

$\Rightarrow \lim\limits_{n\to\infty}\left\{\left(\sqrt{5}+2\right)^n\right\}=\lim\limits_{n\to\infty}1-\left\{\left(\sqrt{5}-2\right)^n\right\}=1$

$\therefore$ Given limit$=1(-1)=-1$

If n is odd, then $\left[\left(\sqrt{5}+2\right)^n\right]$ is even and $f=f'$

$\Rightarrow \lim\limits_{n\to\infty}\left\{\left(\sqrt{5}+2\right)^n\right\}=\lim\limits_{n\to\infty}\left\{\left(\sqrt{5}-2\right)^n\right\}=0$

$\therefore$ Given limit $0\times1=0$

8. f is continuous function in [a, b], g is a continuous function in [b, c]. A function h(x) is defined as

$h(x)=\begin{cases}f(x) \text{ for } x\in[a,b)\\ g(x) \text{ for } x\in(b,c]\end{cases}$, If $f(b)=g(b)$, then

(A) h (x) has a removable discontinuity at x = b
(B) h (x) may or may not be continuous in [a, c]
(C) $h(b^-)=g(b^+)$ and $h(b^+)=f(b^-)$
(D) $h(b^+)=g(b^-)$ and $h(b^-)=f(b^+)$

Ans. AC

Sol. Given f is continuous in [a, b] .....(i)

g is continuous in [b, c] .....(i)

f(b) = g(b) .....(iii)

$\left.\begin{array}{l}h(x)=f(x)\text{ for } x\in[a,b)\\ =f(b)=g(b)\text{ for } x=b\\ =g(x)\text{ for } x\in(b,c]\end{array}\right\}$ .....(iv)

h(x) is continuous in $[a,b)\cup(b,c]$ [Using Eqs. (i) and (ii)]

Also, $f(b^-)=f(b)$ and $g(b^+)=g(b)$ [Using Eqs. (i) and (ii)].....(iv)

$\therefore\ h(b^-)=f(b^-)=f(b)=g(b^+)=h(b^+)$ [Using Eqs.(iv) and (v)]

Now, verify each alternative. Of course $g(b^-)$ and $f(b^+)$ are undefined.

$h(b^-)=f(b^-)=g(b)=g(b^+)=h(b^+)=g(b^+)=g(b)=f(b)=f(b^-)$

Hence, $h(b^-)=h(b^+)=f(b)=g(b)$

and h(b) undefined

9. If $2^{(\cos^{-1}x)^2}\sqrt{(\sin^{-1}y)^4-2(\sin^{-1}y)^2+2}=1$, then the value of (x – y) can be :

(A) $1-\frac{\pi}{2}$ (B) $1+\frac{\pi}{2}$

(C) $1-\sin 1$ (D) $1+\sin 1$

Ans. CD

Sol. $2^{(\cos^{-1}x)^2}\sqrt{(\sin^{-1}y)^4-2(\sin^{-1}y)^2+2}=1$

$2^{(\cos^{-1}x)^2}\sqrt{\left((\sin^{-1}y)^2-1\right)^2+1}=1$

Now $2^{(\cos^{-1}x)^2}\geq 1$ and $\sqrt{\left((\sin^{-1}y)^2-1\right)^2+1}\geq 1$

$\Rightarrow 2^{(\cos^{-1}x)^2}\sqrt{(\sin^{-1}y-1)^2+1}\geq 1$

$\Rightarrow 2^{(\cos^{-1}x)^2}=1$ and $\left((\sin^{-1}y)^2-1\right)^2+1=1$

$\Rightarrow (\cos^{-1}x)^2=0$ and $(\sin^{-1}y)^2=1$

$\Rightarrow x=1$ and $\sin^{-1}y=\pm 1$

$\therefore y=\sin 1$ or $-\sin 1$

$\therefore\ x-y=1+\sin 1$ or $1-\sin 1$

10. Which of the following is periodic ?

(A) $f(x)=\{x\}$,where $\{x\}$ denotes the fractional part of the number x

(B) $f(x)=\cot(x+7)$

(C) $f(x)=1-\frac{\sin^2 x}{1+\cot x}-\frac{\cos^2 x}{1+\tan x}$

(D) f (x) = x + sin x

Ans. ABC

Sol. The function in (a) is periodic with period, 1

The function in (b) is periodic with period π .The function in (c) on simplication is equal to (1/2) sin 2x, which is periodic with period π .The function in (d) is non-periodic as g(x) = x is non-periodic.

11. If 'a' is a real number for which $f(x)=\ln\left(3\cos^{-1}\left(\frac{3x}{7}\right)-\pi\right)$ is defined then the possible value(s) of [a] is (where [.] denotes greatest integer function)

(A) –2 (B) –1

(C) 1 (D) 0

Ans. ABCD

Sol. $3\cos^{-1}\left(\frac{3x}{7}\right)-\pi>0 \Rightarrow \frac{\pi}{3}<\cos^{-1}\left(\frac{3x}{7}\right)<\pi$

$\Rightarrow -\frac{7}{3}<x<\frac{7}{6} \Rightarrow -2.33<x<1.16 \qquad so[x]=-3,-2,-1,0,1$

12. Let $f(x)=\begin{cases}\dfrac{(2^x-1)^2}{\tan x.\ln(1+x)}; x\neq 0\\ \ln 4 \quad ; x=0\end{cases}$

then f (x) is

(A) Continuous at x = 0

(B) Non-removable discontinuity at x = 0

(C) f (x) has a removable discontinuity at x = 0

(D) $\lim\limits_{x\to 0} f(x)$ exists

Ans. CD

Sol. $\lim\limits_{x\to 0} f(x)=\lim\limits_{x\to 0}\left(\frac{2^x-1}{x}\right)^2\cdot\frac{1}{\frac{\tan x}{x}\cdot\frac{\ln(1+x)}{x}}$

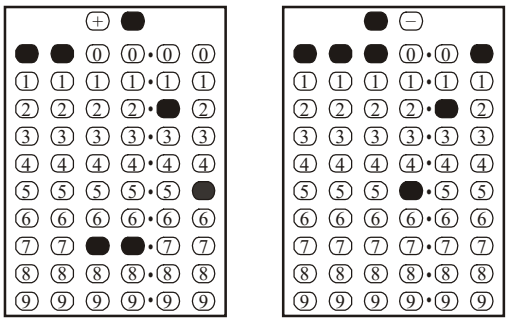
$=\lim\limits_{x\to 0}\left(\frac{2^x-1}{x}\right)^2\cdot\frac{1}{\lim\limits_{x\to 0}\frac{\tan x}{x}\lim\limits_{x\to 0}\log(1+x)^{\frac{1}{x}}}$

$=(\ln 2)^2\cdot\frac{1}{1}\cdot\frac{1}{\ln e}=(\ln 2)^2\neq f(0) \qquad (\because f(0)=\ln 4)$

$\therefore$ $f(x)$ is discontinuous at x = 0. This is removable discontinuity. (if we redefine $f(0)=(\ln 2)^2$, f will be continuous at x = 0)

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : +3 If ONLY the correct numerical value is entered as answer.
  Zero Marks : 0 In all other cases.

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13. Define functions $f_0, f_1, ..., f_n, ...$ to satisfy the following relations : $f_0 = 2, \frac{d}{dx}(f_n) = f_{n-1}$.

If for $n \geq 1, f_n(0) = 0$, compute $\sum_{n=0}^{\infty} f_n\left(\ell n(2019)\right)$

Ans. 4038

Sol. We see by induction that $f_n = 2\frac{x^n}{n!}$. Using the Taylor series of e^x, we have

$$\sum_{n=0}^{\infty} f_n\left(\ell n(2018)\right) = 2\sum_{n=0}^{\infty} \frac{1}{n!}\ell n(2019)^n = 2 \cdot 2019 = 4038$$

14. Let $f(x)$ & $g(x)$ be continuous function for all

$x \in R$. If $\lim_{x\to 3} \frac{f(x)}{x-3} = \lim_{x\to f(3)} \frac{g(x)}{x^2} = 5$ then $\lim_{x\to g(f(3))} \frac{\tan 10x}{x} =$

Ans. 10.00

Sol. Given $\lim_{x\to 3} \frac{f(x)}{x-3} = 5$

$\lim_{x\to 3} f(x) = 0 \Rightarrow f(3) = 0$

$\lim_{x\to f(3)} \frac{g(x)}{x^2} = 5 \Rightarrow \lim_{x\to 0} \frac{g(x)}{x^2} = 5, \quad \Rightarrow g(0) = 0$

So $\lim_{x\to g(f(3))} \frac{\tan 10x}{x} = 10$

15. Let $f(x)=\frac{e^x-e^{-x}}{2}$ and if f (g(x)) = x then $g\left(\frac{e^{1002}-1}{2e^{501}}\right)$ is equal to

Ans. 501

Sol. Let $y=\frac{e^x-e^{-x}}{2}\Rightarrow e^{2x}-1=2ye^x$

Therefore, $t^2-2yt-1=0, t=e^x$

$$t=\frac{2y\pm\sqrt{4y^2+4}}{2}=y\pm\sqrt{y^2+1}$$

$$\Rightarrow x=\ln\left(y+\sqrt{y^2+1}\right)\left(\text{since } e^x>0\right)$$

$$\therefore f^{-1}(x)=g(x)=\ln\left(x+\sqrt{x^2+1}\right)$$

$$g\left(\frac{e^{1002}-1}{2e^{501}}\right)=\ln\left(\frac{e^{1002}-1}{2e^{501}}+\frac{e^{1002}+1}{2e^{501}}\right)$$

$$=\ln e^{1002}=501$$

16. If $2<x^2<3$ then the number of positive roots of $\left\{\frac{1}{x}\right\}=\left\{x^2\right\}$, denotes the fractional part of x, is

Ans. 1

Sol. $2<x^2<3\Rightarrow\sqrt{2}<x<\sqrt{3}\ (\because x>0)$

$$\therefore\left\{\frac{1}{x}\right\}=\frac{1}{x}\text{ and }\left\{x^2\right\}=x^2-2$$

So, the equation becomes

$$\frac{1}{x}=x^2-2\Rightarrow x^3-2x-1=0$$

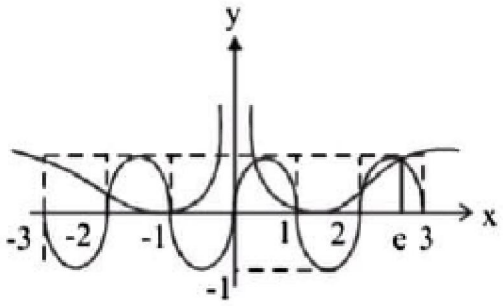
$$\text{or } (x+1)\left(x^2-x-1\right)=0\Rightarrow x=-1,\frac{1\pm\sqrt{5}}{2}$$

$$\therefore x=\frac{1+\sqrt{5}}{2}\ (\because x>0)$$

17. The number of solutions of the equation $\sin \pi x = \left|\log_e |x|\right|$ is

Ans. 6

Sol. The graph of $y = \sin \pi x$ and $y = \left|\log_e |x|\right|$ intersect the 6 points (four positive and two negative roots) as clear from the following figures.



18. Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$

$$265\left(\lim_{h\to 0}\frac{h^4 + 3h^2}{(f(1-h) - f(1))\sin 5h}\right) =$$

Ans. 3

Sol. $$265\left(\lim_{h\to 0}\frac{h^2 + 3}{-\left(\frac{f(1-h)-f(1)}{-h}\right)\left(\frac{\sin 5h}{h}\right)}\right) = -265 \times \frac{3}{f'(1).5} = -\frac{53\times 3}{f'(1)} \quad = \frac{-53\times 3}{-53} = 3$$